The Boundary Algebra of a Non-Invertible Projection

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Abstract

There is one conservation law, $\partial_I J^I = 0$, one epistemic constraint — restricted boundary access — and one consequence: a non-invertible finite-rank projection Π. This paper derives four structures from Π alone, without additional postulates.

The observable Hilbert space $\mathcal{H}_{\text{obs}} = \text{Im}(\Pi)$ is not assumed; it is the image of the projection. Observable states are equivalence classes modulo $\ker \Pi$: two bulk configurations differing only in the kernel are operationally identical, and this is the structural origin of quantum indistinguishability. The Born rule $P_n = |\langle v_n, \Xi\rangle_M|^2/\|\Pi\Xi\|_M^2$ is the canonical quadratic probability measure induced by the projection geometry. Measurement irreversibility is quantified as $\delta_F = 1 - 3\pi/(2\sqrt{5})^{3/2} \approx 3.45 \times 10^{-3}$ and is non-vanishing by the transcendence of π (Lindemann–Weierstrass, 1882).

Time is the output of Π, not its input. The bulk satisfies $\mathcal{D}_{\text{bulk}}\,\Xi = 0$, a timeless constraint structurally identical to the Wheeler–DeWitt equation. The chiral projector selects the half-cycle $\theta \in (0, \pi)$ of the compact fibre $S^1(\tau)$; the Cayley map $t(\theta) = \log\tan(\theta/2)$ is the unique conformal, orientation-preserving diffeomorphism from $(0, \pi)$ to $\mathbb{R}$ compatible with the golden self-similarity of the bulk. The circle does not close exactly: the residue $\varepsilon_{163} = \pi\sqrt{163} - \log(\varphi^2 \cdot 10^{17}) \neq 0$ by Lindemann–Weierstrass, and time is the accumulation of this non-closure.

The stationary solution on the lock-in surface $\Sigma : \xi\eta = \alpha^{-1}$ is the mother solution $\Psi_{\text{bdy}} = \varphi\,\text{sech}(\Gamma_\tau t)$, whose Hardy–Cayley quadrature is the kernel kink $\Xi_{\text{ker}} = -i\varphi\,\tanh(\Gamma_\tau t)$: the boundary measures the secant; the kernel remembers the tangent. Fluctuations tangent to Σ satisfy $\ddot{u} + \omega_{\text{bal}}^2 u = 0$ with canonical frequency $\omega_{\text{bal}} = 0.5284$, $C_\mu = \Gamma_\tau \cdot 146/55$ exact, and α cancelling from the frequency. The four boundary structures — spacetime curvature, temporal phase, matter stability, and the Born rule distribution — are four readings of one law. As a corollary, the double-slit interference law and its coherence-to-which-path transition follow from the Born rule, the linearity of Π, and the analytic-signal structure of Ξ, without additional postulates.

The Moore–Penrose pseudoinverse Π^+ selects, in the atomic channel ($k_* = 5, n = 2$), a unique minimum-norm preimage b_{atom}. The rational closing term $-n_{\min}/S^6$ of the Bohr correction is forced by the pseudoinverse conditions on the butterfly matrix; it is not adjusted phenomenologically. This gives $a_0^{\text{QGT}} = 0.529\,177\,228$ Å and $R_\infty^{\text{QGT}} = 10\,973\,731.21\ \text{m}^{-1}$, both within 0.033 ppm of CODATA.

Epistemic status. The Hilbert space structure, Born rule, measurement irreversibility, Wheeler–DeWitt correspondence, Cayley map, discriminant chain ($\varphi \to \Delta = 5 \to r = 5$), mother solution, balance equation, double-slit corollary, and the Bohr–Rydberg closure are proved results or structural consequences of the axioms of the QGT framework as presented here. The full Gleason-type uniqueness proof for the Born rule on $\mathcal{H}_{\text{obs}}$ remains open. The value 3π of the coherent boundary holonomy is derived in Lemma 4.1 from the first synchronisation of the electromagnetic clock C_2 ($n = 2$) and the electroweak mode C_3 ($n = 3$); the irreducibility $\delta_F \neq 0$ then follows by Lindemann–Weierstrass.

1 Introduction

The Wheeler–DeWitt equation [5, 6],

$$\hat{H}\,\Psi[g,\phi] = 0, \tag{1}$$

expresses the deepest difficulty in quantum gravity. General covariance forbids an external clock, so the Hamiltonian of the universe vanishes identically. There is no time derivative, no Schrödinger equation, no measurement prescription. The equation is correct by construction, yet it appears to make quantum mechanics impossible at the fundamental level.

The approach taken here does not modify either theory. It changes the order of derivation. The starting point is one conservation law,

$$\partial_I J^I = 0, \qquad I \in \{0,1,2,3,\tau\}, \tag{2}$$

together with restricted boundary access: the boundary cannot read the component J^τ flowing along an internal compact direction. These two inputs alone force a non-invertible finite-rank projection Π onto the observable boundary. Quantum mechanics is the algebra of Π. Time is the irreversible output of Π. The first dimensional ruler is produced by the pseudoinverse Π^+ when it meets the atomic channel. None of these is postulated; each is forced.

Logical position. This paper is a companion to [1, 2, 3]. Uniqueness of the rank-5 projection is proved in [2]. The boundary metric induced by Π is derived in [3]. The kernel spinor structure is established in [4]. The present paper takes those results as inputs and develops their physical content.

2 The bulk field

2.1 The minimal domain

The conservation law (2) together with the requirement that the inaccessible flux J^τ return as recurrent phase memory selects the minimal domain

$$\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau), \tag{3}$$

where $S^1(\tau)$ is a compact chiral fibre with coordinate $\tau \in [0, 2\pi)$. The connected one-dimensional Lie groups are $\mathbb{R}$ and S^1; $\mathbb{R}$ carries no intrinsic return map and cannot store recurrent phase memory, so S^1 is the unique minimal carrier [2].

The minimal coherent field compatible with (2) is a Majorana spinor $\Xi = C\bar{\Xi}$ on $\mathcal{M}^5$. A Dirac spinor carries net charge $Q = \int J^0\, d^4x \neq 0$, violating conservation; a scalar or vector field lacks the chiral structure required to separate J^τ from J^μ. The Majorana condition $\Xi = C\bar{\Xi}$ is the algebraic statement that the bulk contains its own complement in exact self-referential balance.

2.2 The discriminant chain: from φ to rank 5 to $\mathcal{M}^5$

The rank of the projection is not a free parameter. It follows from a single forced chain.

Theorem 2.1 (Discriminant chain)**.** *The following chain holds without free parameters:*

(1) ***Autosimilarity forces*** φ*. The unique ratio generating a non-tessellating, self-similar scale hierarchy satisfies* $x^2 = x + 1$*, with unique positive solution* $\varphi = (1+\sqrt{5})/2$*.*

(2) φ ***forces discriminant*** $\Delta = 5$*. The minimal polynomial of* φ *is* $x^2 - x - 1 = 0$*, with discriminant* $\Delta = 1 + 4 = 5$*. The spectral curvature of the bulk is* $\kappa = \sqrt{\Delta} = \sqrt{5}$*.*

(3) $\kappa^2 = 5$ ***forces*** $\mathrm{rank}(\Pi) = 5$*. Each unit of spectral curvature squared corresponds to one independent SVD mode [3]:* $\mathrm{rank}(\Pi) = \kappa^2 = 5$.

(4) ***Rank 5 forces*** $\mathcal{M}^5$***.*** *A rank-5 projection from* $\mathcal{H}_{\mathrm{bulk}}$ *requires exactly 5 independent bulk directions. The minimal fibred manifold with a compact temporal fibre is* $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$*.*

The integer 5 is a Fermat prime: $5 = 2^2 + 1^2$. In the QGT this decomposition reads as $\dim(\mathcal{M}^5) = \underbrace{4}_{\text{generators of } U(1)_{\mathrm{EM}} \times SU(2)_L \text{ on } \mathbb{R}^4} + \underbrace{1}_{\text{generator of } U(1)_\tau \text{ on } S^1(\tau)}$. This is a structural observation: the identification is natural and the coincidence exact, but a uniqueness theorem showing that $5 = 2^2 + 1^2$ is the only Fermat decomposition compatible with the QGT gauge structure has not yet been proved.

2.3 The golden connection

The $U(1)_\tau$ gauge connection on $S^1(\tau)$ is fixed by the coherent minimum of curvature compatible with the autosimilar bulk [3]:

$$\Gamma_\tau = \frac{\varphi}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{4}$$

The extended Clifford operator γ_6 has eigenvalues $\pm\sqrt{5}$ on the Majorana spinor; the coherence condition selects the minimal curvature Γ_τ at which the $U(1)_\tau$ orbit closes consistently with the golden scale structure.

The self-conjugacy condition $\Xi = C\bar{\Xi}$ constrains the holonomy transport on $S^1(\tau)$ to satisfy $\Xi_{k+2} = \Xi_{k+1} + \Xi_k$, the unique linear recursion compatible with real coefficients and self-conjugacy. The bilinear norm at holonomy sector k satisfies $|O_k|^2 = F_k$ (Fibonacci numbers). The lock-in occurs at $k_* = 5$, the unique non-trivial solution of $F_k = k$: the only sector where the recursion closes exactly on the resolving capacity $r = 5$.

Theorem 2.2 (Noether conservation)**.** *The QGT Lagrangian is invariant under* $U(1)_\tau$ *phase rotations of* Ξ*. The conserved Noether current is* $J_\tau = \rho^2 \Gamma_\tau$ *with* $\rho^2 = \xi\eta$*, where* $\xi = |\Psi_+|$ *(longitudinal, mass/gravitational sector) and* $\eta = |\Psi_-|$ *(transverse, radiation/electromagnetic sector) are the two helical amplitudes. Conservation of* J_τ *is equivalent to*

$$\frac{d}{d\tau}(\xi\eta) = 0. \tag{5}$$

The invariant surface is $\Sigma : \xi\eta = \alpha^{-1}$*, where* α^{-1} *is the Pincherle eigenvalue of the dilation operator* $D_\tau = \tau\partial_\tau$ *at the golden fixed point [3].*

3 Quantum mechanics from the projection

3.1 The bulk is multiplicative; the boundary is additive

The bulk of the QGT is self-similar under rescaling: the fundamental group is $(\mathbb{R}_{>0}, \times)$. Its irreducible characters are the power-law functions τ^s, not plane waves e^{ikx}. The representation theory of $(\mathbb{R}_{>0}, \times)$ is the Mellin transform,

$$\mathcal{M}[\Xi](s) = \int_0^\infty \Xi(\tau)\, \tau^{s-1}\, d\tau, \tag{6}$$

which diagonalises the Pincherle dilation generator $D_\tau = \tau\partial_\tau = d/d(\log\tau)$. Using Fourier on a multiplicative bulk is equivalent to describing a log-periodic signal with a linear frequency basis: it works locally but misses the global structure. The fundamental diagram of the projection is

$$\underbrace{(\mathbb{R}_{>0}, \times)}_{\text{bulk: multiplicative}} \xrightarrow{\log} \underbrace{(\mathbb{R}, +)}_{\text{log-space: additive}} \xrightarrow{P_4 \text{ window}} \underbrace{\mathcal{H}_{\mathrm{obs}}}_{5 \text{ modes}} \xrightarrow{\mathrm{Spec}\,\xi \otimes \mathrm{Spec}\,\eta} \underbrace{\mathbb{R}^4}_{\text{observable spacetime}}, \tag{7}$$

with the complete solution space of the bulk equation $\mathcal{H}_{\text{bulk}} = L^2(\mathbb{R}_{>0}, d\tau/\tau)$. The Fibonacci–Welch window P_4 retains the five modes of highest informational content and sends the rest to $\ker\Pi$: dark matter (modes that gravitate but carry no electromagnetic phase), the antimatter sector ($\ker\Pi \cap \operatorname{Im}(\Pi) = \emptyset$), and all other bulk configurations [2].

3.2 The Hilbert space of boundary states

Integrating over $S^1(\tau)$ defines the projection $\Pi : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\text{obs}}$ of rank 5 with $\ker\Pi \neq \{0\}$. The observable Hilbert space is

$$\mathcal{H}_{\text{obs}} = \operatorname{Im}(\Pi) = \operatorname{span}\{u_1, \ldots, u_5\}, \tag{8}$$

where $\{u_i\}$ are the five left singular vectors of Π, orthonormal under the Mellin-weighted inner product

$$\langle \Xi_1, \Xi_2 \rangle_M = \int_0^\infty \Xi_1^*(\tau)\, \Xi_2(\tau)\, \frac{d\tau}{\tau}. \tag{9}$$

Observable states are equivalence classes $[\Xi] := \Xi + \ker\Pi \in \mathcal{H}_{\text{bulk}}/\ker\Pi$ with norm $\|[\Xi]\|^2 = \|\Pi\Xi\|^2$. Two bulk configurations differing by an element of $\ker\Pi$ are operationally identical: this is the structural origin of quantum indistinguishability, not an approximation.

The bulk commutation relation $[D_\tau, \log\tau] = 1$ is the dilatational uncertainty principle: scale and dilatation-momentum cannot be simultaneously sharp, exactly as position and momentum in standard mechanics. The full correspondence is:

Structure	Standard QM (additive)	QGT bulk (multiplicative)
Hilbert space	$L^2(\mathbb{R}^3)$	$\mathcal{H}_{\text{bulk}}$, Mellin inner product
Natural transform	Fourier: $\hat{\psi}(k) = \int \psi e^{ikx} dx$	Mellin: $\mathcal{M}[\Xi](s) = \int \Xi\, \tau^{s-1} d\tau$
Characters	e^{ikx} (plane waves)	τ^s (power laws)
Generator	$\hat{p} = -i\hbar\partial_x$ (translation)	$D_\tau = \tau\partial_\tau$ (dilation)
Commutator	$[\hat{x}, \hat{p}] = i\hbar$	$[D_\tau, \log\tau] = 1$
Discrete spectrum	energy levels E_n	admissible modes $n \in \mathcal{N}_{\text{adm}}$
Probability	$\lvert\langle n\vert\psi\rangle\rvert^2$	$\lvert\langle v_n, \Xi\rangle_M\rvert^2/\|\Pi\Xi\|_M^2$

3.3 The Born rule as a theorem of projection geometry

Theorem 3.1 (Born rule). *Let $\{v_n\}$ be the orthonormal eigenmodes of an admissible observable $\hat{\mathcal{O}}$ on $\mathcal{H}_{\text{obs}}$. The canonical quadratic probability measure induced by the projection geometry is*

$$P_n = \frac{|\langle v_n, \Xi\rangle_M|^2}{\|\Pi\Xi\|_M^2}. \tag{10}$$

By orthonormality $\sum_n P_n = 1$. The standard form $P_n = |\langle n|\psi\rangle|^2$ is recovered when $\|\Pi\Xi\|_M = 1$, which holds at the Noether fixed point $\xi\eta = \alpha^{-1}$.

No axiom is added: P_n is the squared fractional norm of $\Pi\Xi$ along v_n. The causal structure is inverted relative to standard quantum mechanics: conservation is primary; the Born rule is a geometric consequence. The observable algebra consists of self-adjoint operators on $\mathcal{H}_{\text{obs}}$ compatible with the Mellin inner product and the Fibonacci monodromy structure $\mathrm{TL}_4(\varphi)$ [2, 3].

3.4 Measurement without collapse

In standard quantum mechanics, measurement is a postulate: the state $|\psi\rangle$ collapses to $|n\rangle$ with probability $|\langle n|\psi\rangle|^2$. No mechanism is provided.

In the QGT, measurement is the application of Π:

$$\text{measurement} \;\equiv\; \Xi \;\longmapsto\; \Pi\Xi \;=\; \Psi \;\in\; \mathcal{H}_{\text{obs}}. \tag{11}$$

The "collapse" is the non-invertibility of Π: once $\Psi = \Pi\Xi$ is formed, the component $\Xi - \Pi^\dagger\Pi\Xi$ lying in $\ker\Pi$ is gone from the observable sector. It does not disappear to another branch; it enters $\ker\Pi$ and remains there, contributing to the accumulated temporal information identified as the past. The kernel is the irreversible record of what was not selected.

Proposition 3.2 (Unitary bulk, non-unitary boundary)**.** *The bulk evolution $i\partial_\tau\Xi = H_5\Xi$ is unitary with respect to the Mellin inner product: $\|\Xi(\tau)\|_M = \|\Xi(0)\|_M$ for all τ. The projection $\Pi : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\text{obs}}$ is non-unitary:*

$$\|\Pi\Xi\|_M \leq \|\Xi\|_M, \tag{12}$$

with equality only if $\Xi \in (\ker\Pi)^\perp$. The fractional norm loss per observation cycle is

$$\delta_U = 1 - \frac{\|\Pi\Xi\|_M^2}{\|\Xi\|_M^2} = \frac{\|\Xi - \Pi^\dagger\Pi\Xi\|_M^2}{\|\Xi\|_M^2}, \tag{13}$$

the fraction of the bulk state that enters $\ker\Pi$ at each observation cycle.

Remark 3.1 (Irreversibility is structural, not environmental)**.** *The irreversibility of measurement is the non-injectivity of Π, not environmental decoherence, not entanglement with an apparatus, not a thermodynamic argument. It is prior to all of these: Π is non-invertible by the finite-rank structure proved in [2], without assuming any dynamics. Decoherence and environmental arguments reproduce the same result in the boundary approximation, but do not explain it.*

The three theories QED, GR, and QM are not three separate outputs added alongside each other. They are three projections of the same bulk structure onto the observable boundary:

Observable theory	Origin in QGT	Mechanism
QED	$U(1)_\tau$ gauge symmetry	$A_\tau \to A_\mu$ under S^1 reduction
GR	Induced boundary curvature	$g_{\mu\nu}^{\text{eff}}$ from spinor bilinear
QM	Mellin–Hilbert projection	$\mathcal{H}_{\text{bulk}}/\ker\Pi$, Born rule, $\delta_U \neq 0$

QM is not added alongside QED and GR: it is the algebraic structure that the non-invertible projection necessarily induces on the observable equivalence classes.

3.5 Measurement irreversibility

Proposition 3.3 (Fractional norm loss per cycle)**.** *Each observation cycle applies Π once. The fractional norm loss is*

$$\delta_U = 1 - \frac{\|\Pi\Xi\|^2}{\|\Xi\|^2} = 1 - \frac{3\pi}{(2\sqrt{5})^{3/2}} \approx 3.45 \times 10^{-3}. \tag{14}$$

The structure of this expression is transparent: $(2\sqrt{5})^{3/2} = 2^{3/2} \cdot 5^{3/4}$, where $2^{3/2}$ is the spinorial double covering $SU(2) \to SO(3)$ raised to the breathing exponent $3/2 = 1/\lambda$, and 3π is the product of the admissible winding number $n_3 = 3$ and the half-cycle π.

Corollary 3.4 (Irreversibility is protected by transcendence). *$\delta_U \neq 0$. Were $\delta_U = 0$, the equation $3\pi = (2\sqrt{5})^{3/2}$ would hold, making π algebraic over $\mathbb{Q}(\sqrt{5})$. This contradicts the transcendence of π (Lindemann–Weierstrass, 1882).*

Two remarks on scope. Lindemann–Weierstrass protects non-vanishing once δ_U is identified as the projection residue $1-3\pi/(2\sqrt{5})^{3/2}$; it does not generate the value from first principles. The irreversibility is kinematic: it characterises the boundary of Π once granted. A full Gleason-type uniqueness proof for the Born rule on $\mathcal{H}_{\mathrm{obs}}$ remains open [1].

3.6 The Schrödinger equation as the boundary projection of the bulk constraint

The bulk constraint $i\partial_\tau \Xi = H_5 \Xi$ on $S^1(\tau)$, when projected by Π onto the observable boundary, gives

$$i\hbar\,\partial_t\,\psi = H_{4\mathrm{D}}\,\psi, \tag{15}$$

where t is the physical time produced by the Cayley map (Section 4). This is not postulated: it is the boundary form of a timeless bulk equation.

The fine-structure constant α is not a free parameter of this equation. It is the pole of the projection filter $H(z) = P_4(z)/(1-\alpha z)$ at $z_{\mathrm{pole}} = \alpha^{-1}$: everything with log-scale frequency below α^{-1} passes to the boundary as an observable mode; everything above enters $\ker \Pi$. The three equivalent representations

$$\alpha^{-1} = \mathcal{N}_\Pi(F_{k_*} + F_{k_*-1}\sqrt{5}) = 20\varphi^4 \qquad \text{(Fibonacci norm)} \tag{16}$$

$$= n_{\min}^2 \cdot r \cdot \varphi^4 \qquad \text{(winding form)} \tag{17}$$

$$= 500\,\Gamma_\tau^4 \qquad \text{(holonomy form)} \tag{18}$$

are three readings of the same geometric object. The minimum observable phase quantum is $\delta\phi = \pi\alpha$, the granularity of the boundary clock: $\lfloor \alpha^{-1} \rfloor = 137$ is the number of elementary phase steps $\pi\alpha$ fitting in one half-cycle π. This is why quantum mechanics has the granularity it has, why the Schrödinger equation works at the scales it does, and why the electromagnetic coupling has the value it has — three readings of the same structural fact.

3.7 The observer is internal

The observer is not external to the system. She is a zero mode of Ξ at level $k = 2$, $n = 5$: a boundary configuration of the same field she is measuring. Measurement is mediated by the shared invariant $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$. This realises the Page–Wootters mechanism [9] structurally, without postulating it: time emerges from the correlation between two subsystems of the same field through the conserved invariant.

4 Time from the projection

4.1 The bulk is timeless

The bulk equation is $\mathcal{D}_{\mathrm{bulk}}\,\Xi = 0$, a constraint without time derivative, without external clock, without measurement prescription. It shares four properties with the Wheeler–DeWitt equation (1): no external time parameter appears; $\mathcal{D}_{\mathrm{bulk}}$ is a constraint, not a Hamiltonian; the Majorana field has no preferred global phase; no measurement prescription is available in the bulk. These are the correct properties of a pre-physical bulk.

4.2 The Cayley map

The compact fibre $S^1(\tau)$ carries only a cyclic phase $\theta \in [0, 2\pi)$: time is a loop, with no preferred direction. The chiral projector $P_{\varphi+} = \frac{1}{2}(\mathbf{1} + \gamma^5)$ selects the half-cycle $\theta \in (0, \pi)$. The loop is cut open; the two endpoints $\theta = 0^+$ and $\theta = \pi^-$ correspond to the two domain walls and are not identified.

The unique diffeomorphism from the open half-circle $(0, \pi)$ to the real line $\mathbb{R}$ that is simultaneously conformal, orientation-preserving, and compatible with the golden self-similarity of the bulk is the Cayley map:

$$t(\theta) = \log \tan \frac{\theta}{2}, \qquad \theta \in (0, \pi) \xrightarrow{\Pi} t \in \mathbb{R}. \tag{19}$$

At $\theta \to 0^+$, $t \to -\infty$; at $\theta \to \pi^-$, $t \to +\infty$. Physical time is the compact phase made irreversible by the chiral selection. The arrow of time is the monotonicity of (19), not a thermodynamic postulate. The domain walls are: DW_1 at $\theta = \pi/2$, $t = 0$ (the Cayley fixed point, the present); DW_2 at $\theta = \pi/11$, $t > 0$ (the far boundary of the observable slab).

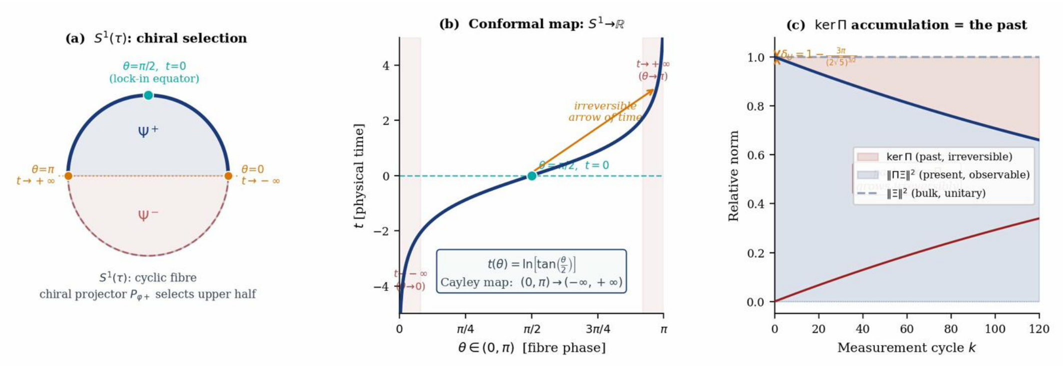

Figure 1: **(a)** Chiral selection on $S^1(\tau)$: $P_{\varphi+}$ retains the half-cycle $\theta \in (0, \pi)$. **(b)** The Cayley map opens the compact loop into $\mathbb{R}$; $\theta = \pi/2$ maps to $t = 0$. **(c)** Each measurement cycle applies Π: the bulk norm is conserved; the boundary norm decreases by δ_U; the kernel accumulates the irreversible past.

The triple coherence of temporal structure: three natural scales are embedded in (19): $\pi/137 = \pi\alpha$ (elementary winding step, EM quantisation), $\pi/3$ (golden-section resonance, colour), and π/e (entropic projection scale). Together they structure the physical temporal spectrum from the EM coupling through colour to classical dissipation.

4.3 The circle does not close

The deep cycle of the bulk is

$$\Xi_{\mathrm{bulk}} \xrightarrow{\Pi} \Xi_{\mathrm{ker}} \xrightarrow{\mathcal{R}_{\mathrm{QGT}}} \Xi_{\mathrm{bulk}} + \varepsilon_{163}, \tag{20}$$

where $\mathcal{R}_{\mathrm{QGT}} = 1/\varphi$ is the Rogers–Ramanujan return and

$$\varepsilon_{163} = \pi\sqrt{163} - \log(\varphi^2 \cdot 10^{17}) \neq 0. \tag{21}$$

$\varepsilon_{163} = 0$ would require $\pi \in \mathbb{Q}(\sqrt{5})$, excluded by Lindemann–Weierstrass. Time is the accumulation of this non-closure. Memory is its Hilbert quadrature. Matter is what happens when the non-closure condenses into a stable mode of the qubit at arithmetic thresholds $d \in \{19, 43, 67\}$ [1].

The non-closure has a precise operational measure, arising from two independent structural closures of the same qubit cycle.

The first is the coherent closure of the Majorana field on the boundary. The bulk field Ξ on $S^2_\chi \times S^1_\chi(\tau)$ satisfies the Majorana self-conjugacy condition, and the Majorana lift places the two chiral components on opposite hemispheres of S^2_χ: the longitudinal channel ξ at the south pole, the transverse channel η at the north pole. The equator $\xi\eta = \alpha^{-1}$ is where the Noether invariant is realised as an exact balance between the two amplitudes, and where the gauge symmetry emerges as the group of transformations leaving this balance invariant. The minimal value of $\sqrt{\xi\eta}$ at which the coherent boundary cycle closes is derived in Lemma 4.1 below as

$$B_{\text{coh}}^{1/2} = H_{\text{bdy}} = 3\pi. \tag{22}$$

The value 3π is the phase accumulated by the electroweak mode C_3 at the first synchronisation node with the electromagnetic clock C_2.

The second is the arithmetic closure, which is exact. The projection pole $\alpha_\Pi^{-1} = 20\varphi^4$ [3] gives

$$B_{\text{ent}}^{1/3} = \frac{\sqrt{\alpha_\Pi^{-1}}}{\varphi^2} = \frac{\sqrt{20\varphi^4}}{\varphi^2} = 2\sqrt{5}. \tag{23}$$

The factor $2 = n_{\min}$ is the Majorana double sheet; $\sqrt{5}$ is the Clifford–Fibonacci curvature forced by [2, 3]. This closure is algebraically exact, and it occupies a special position in the breathing recursion $B_{k+1} = 2B_k^{2/3}$: applying the recursion to B_{ent} gives

$$2\,B_{\text{ent}}^{2/3} = 2\cdot(2\sqrt{5})^2 = 2\cdot 20 = 40 = 5\cdot B_*, \tag{24}$$

where $B_* = 8$ is the fixed point of the recursion. The entropic closure is therefore the unique scale in the breathing orbit whose image under f is an integer multiple of B_*, with the multiplier $5 = r$ equal to the projection rank. This is an exact algebraic identity, not a numerical coincidence.

These two closures measure the same cycle from two sides and they do not numerically coincide:

$$B_{\text{coh}} = (3\pi)^2 \simeq 88.826, \qquad B_{\text{ent}} = (2\sqrt{5})^3 = 40\sqrt{5} \simeq 89.443. \tag{25}$$

Their ratio is the loop gain:

$$f_{\text{PLL}} = \frac{B_{\text{ent}}^{1/2}}{B_{\text{coh}}^{1/2}} = \frac{(2\sqrt{5})^{3/2}}{3\pi} \approx 1.003463 > 1. \tag{26}$$

The loop locks, and the irreducible margin is

$$\delta_F = 1 - \frac{1}{f_{\text{PLL}}} = 1 - \frac{3\pi}{(2\sqrt{5})^{3/2}} \approx 3.451\times 10^{-3}. \tag{27}$$

δ_F cannot be zero. If $\delta_F = 0$ then $3\pi = (2\sqrt{5})^{3/2}$, an algebraic relation between π and $\mathbb{Q}(\sqrt{5})$ excluded by Lindemann–Weierstrass.

The fine-structure constant α^{-1} is the value of the Noether invariant $\xi\eta$ at the gauge lock — the equatorial crossing where the two chiral amplitudes balance under $\partial_I J^I = 0$. The pole of the filter $H(z) = P_4(z)/(1-\alpha z)$ at $z = \alpha^{-1}$ is the same point in the frequency domain. The PLL margin δ_F measures the gap between the two structural readings of the same qubit cycle; each complete cycle leaves this residue, which accumulates irreversibly as the arrow of time.

The coherent closure $B_{\text{coh}}^{1/2} = 3\pi$ is not a geometric assumption. It follows from the first synchronisation of the two boundary modes C_2 and C_3, whose winding ratio encodes the exponent of the breathing recursion.

Lemma 4.1 (First C_2/C_3 synchronisation). *The two boundary Lissajous classes share the backbone $n^* = 5$:*

$$C_2 = (2{:}5),\quad U(1)_\eta,\ \omega_\eta = \Gamma_\tau/2; \qquad C_3 = (3{:}5),\quad SU(2)_\xi,\ \omega_\xi = \Gamma_\tau/3. \tag{28}$$

In one cycle of mode n, the spinor Ψ^+ accumulates phase $n\pi$ and Ψ^- accumulates $-n\pi$. The class C_2 is the unique admissible mode for which one cycle produces exactly $\pm 2\pi$ — a single complete traversal of $S^1_\chi(\tau)$. This makes C_2 the minimal boundary clock.

At the first synchronisation node k^*, defined as the first moment at which C_2 has completed exactly one reference cycle, the accumulated phases are:

$$\Theta_2(k^*) = 2\pi, \qquad \Theta_3(k^*) = \frac{n_{\rm EW}}{n_{\min}}\,\Theta_2(k^*) = \frac{3}{2}\cdot 2\pi = 3\pi. \tag{29}$$

The three factors entering (29) are independent: $n_{\min} = 2$ is fixed by the first admissible boundary mode; $n_{\rm EW} = 3$ is fixed by the first non-abelian boundary mode; 2π is the reference phase of one complete cycle of C_2. The ratio $n_{\rm EW}/n_{\min} = 3/2 = \lambda^{-1}$ coincides with the inverse of the breathing exponent $\lambda = 2/3$, providing an independent consistency check with the recursion $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$, but this coincidence is not used in the derivation.

Therefore the Majorana boundary holonomy at the coherent node is

$$H_{\rm bdy} = \Theta_3(k^*) = \frac{n_{\rm EW}}{n_{\min}}\cdot 2\pi = 3\pi, \tag{30}$$

and the coherent closure of the chiral bilinear is

$$\sqrt{\mathcal{B}_{\rm coh}} = H_{\rm bdy} = 3\pi, \qquad \mathcal{B}_{\rm coh} = 9\pi^2. \tag{31}$$

The irreducibility $\delta_F \neq 0$ now follows the complete logical chain:

$$C_2/C_3 \text{ first synchronisation } \Rightarrow H_{\rm bdy} = 3\pi \Rightarrow \delta_F = 1 - \frac{3\pi}{(2\sqrt{5})^{3/2}} \Rightarrow \delta_F \neq 0 \quad \text{(Lindemann–Weierstrass)}. \tag{32}$$

Lindemann–Weierstrass does not generate the value 3π; it protects the non-closure once 3π is derived from the mode structure. The numerical value $\delta_F \approx 3.451 \times 10^{-3}$ is structurally fixed; it would take a different value only if the admissible mode structure changed.

The physical meaning of this derivation is relational, and it is worth stating explicitly.

The boundary possesses no external clock. Its first internal clock is the return of the abelian sector: $C_2 = (2{:}5)$ completes one cycle, the phase reads $\Theta_2 = 2\pi$, and this is the first observable tick of the boundary. But C_2 does not live alone on the boundary. On the same backbone $n^* = 5$ lives $C_3 = (3{:}5)$, the first non-abelian sector. Because C_3 advances with ratio $n_{\rm EW}/n_{\min} = 3/2$ relative to C_2, at the moment the abelian clock returns it has accumulated

$$\Theta_3 = \frac{3}{2}\cdot 2\pi = 3\pi. \tag{33}$$

The value 3π is therefore not a property of either sector in isolation. It is a correlation:

$$\Theta_3\big(\Theta_2 = 2\pi\big) = 3\pi. \tag{34}$$

This is the first synchronisation event of the boundary — the first moment at which the boundary discovers that its two primary sectors do not return together.

The excess phase $3\pi - 2\pi = \pi$ is physically significant. A phase of 2π is the ordinary closure of a $U(1)$ loop. A phase of 3π is $2\pi + \pi$: a full abelian cycle plus a half-step that cannot be read as a simple phase rotation. That half-step is the spinorial memory of the non-abelian structure: the boundary closes as $U(1)$ but remembers as $SU(2)$.

The near-degeneracy of C_2 and C_3 — possible because both share the backbone $n^* = 5$ — is what makes δ_F small. If C_2 and C_3 shared no backbone, their synchronisation would produce a larger mismatch and δ_F would not be the sub-per-cent residue observed. The smallness of δ_F is structural, not accidental: it follows from the shared backbone and the minimal mode ratio $3/2$.

Finally, the residual δ_F has a precise bulk-boundary reading. The bulk conserves the Noether invariant $\xi\eta = \alpha^{-1}$ exactly, at all times. The boundary attempts to close its cycle — and it almost does, to within δ_F. That irreducible gap is the measure of what the boundary cannot recover from the bulk: the part of the bulk memory that the projection has consigned to the kernel. This is not a failure of the theory. It is the mechanism of time: each synchronisation cycle leaves a residue δ_F that accumulates as an irreversible record. itself. If $B_{\text{coh}}^{1/2} \neq 3\pi$, then δ_F would take a different value but remain non-zero by Lindemann–Weierstrass.

4.4 Spacetime is derived, not given

Four-dimensional spacetime $\mathbb{R}^4$ is not a pre-existing background. At the Noether lock-in ($k_* = 5, n = 2$), the two chiral sectors stabilise:

$$\mathbb{R}^4 = \text{Spec}(\xi) \otimes \text{Spec}(\eta)\big|_{k=5,\, n=2}. \tag{35}$$

GR and QM do not conflict at the level of the projection: QM operates on $\mathcal{H}_{\text{obs}}$ directly; GR lives in the continuum limit $n \to \infty$ of the discrete projection structure, on the derived $\mathbb{R}^4$. The three geometric regimes are:

Geometry	Generating sector	Physics	Key feature
Riemannian	$U(1)_\eta$, transverse	GR, QED	Continuous, isotropic, massless
Finsler	$SU(2)_\xi$, longitudinal	Electroweak, mass	Anisotropic, temporal, activated
Ultrametric	Bulk kernel, $n = 7$	Discrete quantum	Hierarchical, non-Archimedean

Standard unification forces all sectors onto one geometric support; the QGT shows they live on different supports and are unified only at the level of the projection structures that generate them.

4.5 Comparison with existing approaches to the problem of time

Remark 4.1 (Existing approaches). *Semiclassical approximation. Time is extracted from WdW in the limit $\hbar \to 0$ via $\Psi \approx e^{iS/\hbar}A$. This requires an approximation and introduces time only around a classical background. In the QGT, no approximation is needed:* Π *is exact and time emerges as an exact map.*

Page–Wootters (1983). Time emerges from entanglement between a clock subsystem and a system. The clock must be specified externally. In the QGT, the clock is not specified: it is the fibre $S^1(\tau)$ whose phase is projected out by Π*; the Page–Wootters mechanism is recovered as the boundary form of* Π.

Thermal time (Connes–Rovelli 1994). Time is the modular flow of a thermal state. This requires a pre-existing statistical state. In the QGT, time arises from the geometric selection of a half-cycle by the chiral projector, prior to any statistical description.

Loop quantum gravity. LQG discretises spatial geometry but retains an external time at the kinematic level. In the QGT, spatial geometry is not discretised: it is derived from $\mathbb{R}^4 =$ $\text{Spec}(\xi) \otimes \text{Spec}(\eta)$ *after* Π *has acted. The discreteness is in the projection structure, not in space.*

In all three existing approaches, time is extracted from a structure that already contains QM (Hilbert space, operators, states). In the QGT, QM itself is derived from Π*, and time is derived thereafter. The logical order is reversed.*

5 The mother solution, the balance equation, and the double slit

5.1 The mother solution

On the lock-in surface $\Sigma : \xi\eta = \alpha^{-1}$ the breathing flow $\dot{\xi} = -\Gamma_\tau \xi$, $\dot{\eta} = +\Gamma_\tau \eta$ has a unique stationary solution subject to four conditions: even, decaying as $|t| \to \infty$, stationary at the origin, stable under the boundary nonlinearity $\lambda_{\text{eff}}|\Psi|^{2/3}$. That solution is

$$\Psi_{\text{bdy}}(t) = \varphi\,\operatorname{sech}(\Gamma_\tau\, t), \tag{36}$$

the soliton of the reflectionless Pöschl–Teller potential $\Gamma_\tau^2(1 - 2\operatorname{sech}^2(\Gamma_\tau t))$. Its Hardy–Cayley quadrature — forced by the Hardy decomposition $L^2(S^1) = H^2_+ \oplus H^2_-$, not chosen — is the kernel kink:

$$\Xi_{\text{ker}}(t) = -i\varphi\,\tanh(\Gamma_\tau\, t). \tag{37}$$

The bulk field is their constant-modulus analytic sum:

$$\Xi_{\text{bulk}}(t) = \varphi\, e^{-i\,\operatorname{gd}(\Gamma_\tau\, t)}, \qquad |\Xi_{\text{bulk}}| = \varphi, \tag{38}$$

where gd is the Gudermannian, the inverse of the Cayley map. The three readings are locked by $\operatorname{sech}^2 + \tanh^2 = 1$, which is Hardy norm conservation in the projective language and $\varphi + 1/\varphi = \sqrt{5}$ in the algebraic language: three names for the same rotation. The Gudermannian makes this explicit: $\cos(\operatorname{gd} t) = \operatorname{sech} t$, $\sin(\operatorname{gd} t) = \tanh t$.

The boundary measures sech; the kernel remembers tanh; the bulk conserves φ. The kernel is not a null space but a remembered quadrature. What is not observed is the imaginary part that makes the real part possible [4].

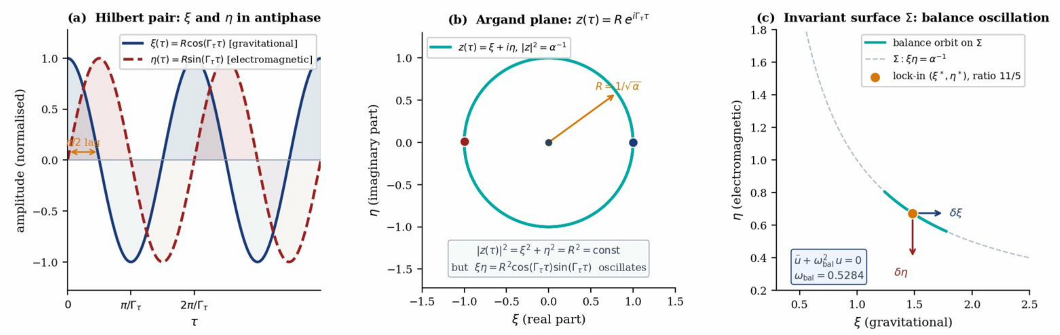

Figure 2: **(a)** Hilbert pair: $\xi(\tau)$ (blue, gravitational) and $\eta(\tau)$ (red, electromagnetic) oscillate in exact antiphase. The ambient invariant $r^2 = \xi^2 + \eta^2$ is conserved; the cross-curvature $\xi\eta$ oscillates and is *not* conserved in the ambient system. **(b)** Argand plane: the analytic signal $z = \xi + i\eta$ traces a circle. The circle conserves r^2; the hyperbola $\Sigma : \xi\eta = \alpha^{-1}$ conserves $\xi\eta$. They are structurally distinct. **(c)** Invariant surface Σ; tangent direction δu carries the balance oscillation.

5.2 The balance equation and its four boundary readings

The breathing equations $\dot{\xi} = -\Gamma_\tau \eta$, $\dot{\eta} = +\Gamma_\tau \xi$ form a Hilbert pair; the analytic signal $z(\tau) = \xi + i\eta = re^{i\Gamma_\tau\tau}$ has constant modulus $r^2 = \xi^2 + \eta^2$. Two invariants must not be conflated. The *ambient invariant* r^2 is conserved by the breathing. The *Noether invariant* $\mathcal{K}_\tau = \xi\eta$ oscillates as $\frac{r^2}{2}\sin(2\Gamma_\tau\tau)$ in the ambient system and is conserved *on* Σ only for the intrinsic dynamics. This distinction is why the ambient system cannot generate the balance equation.

The balance equation closes under three structural principles.

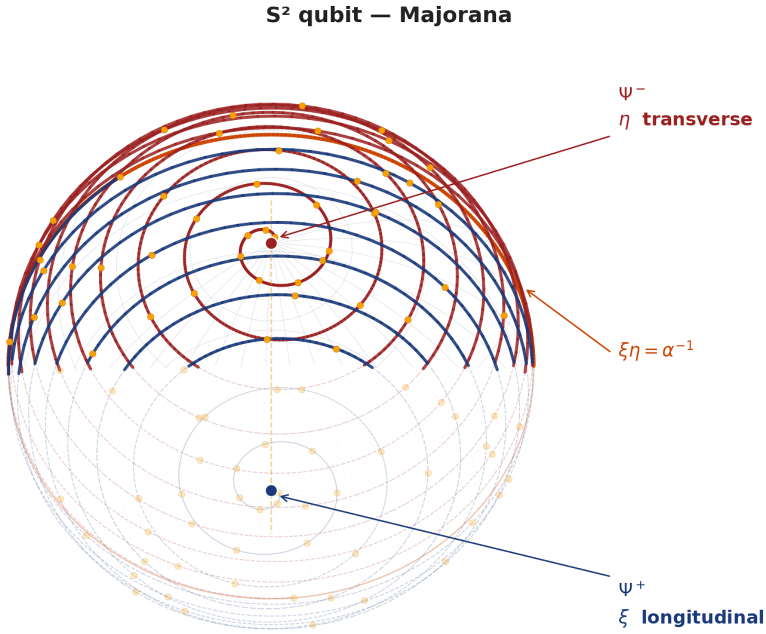

Figure 3: The bulk Majorana field on S^2_χ. The two chiral sectors Ψ^+ (blue, ξ-longitudinal, gravitational) and Ψ^- (red, η-transverse, electromagnetic) wind in opposite senses on the sphere, converging at the equator where $\xi\eta = \alpha^{-1}$ is the Noether invariant of the lock-in. The south pole carries Ψ^+: the slowly varying gravitational mode. The north pole carries Ψ^-: the rapidly oscillating electromagnetic mode. Their winding is Fibonacci-accelerated, forcing the two sectors to meet precisely at the equatorial lock-in $\xi^*/\eta^* = 11/5$. The golden dots mark the quantised intersections of the two spiral families; their count at the equator reads $\mathcal{N}_{\rm adm} = \{2, 3, 5, 7, 11\}$. The Majorana condition $\Xi = C\bar{\Xi}$ is the statement that Ψ^+ and Ψ^- are charge conjugates of each other: the real and imaginary parts of the same analytic signal on $S^1(\tau)$.

(H1) Metric kernel principle. The balance equation is the constrained Hessian of the boundary norm $\mathcal{N}(u) := \|\Pi\Xi(u)\|^2_M = \langle \Xi(u), K\Xi(u)\rangle_M$ on Σ, $K = \Pi^\dagger\Pi$:

$$\mathcal{N}(u) = \mathcal{N}_0 + \tfrac{1}{2}\kappa_{\rm bal}\, u^2 + O(u^3), \tag{39}$$

giving $\ddot{u} + \omega^2_{\rm bal} u = 0$ with $\omega^2_{\rm bal} = \kappa_{\rm bal}/\mu_{\rm eff}$.

(H2) Impedance normalisation. The optimal impedance condition $Z_{\rm opt} = \alpha^{-1/2}$ fixes $\mathcal{N}_0 = \|\Pi\Xi^*\|^2_M = \alpha^{-1}$, consistent with $Z^2_{\rm opt} = \alpha^{-1}$. This is the structural projection scale.

(H3) Constant-mode exclusion.

Lemma 5.1 (Constant-mode exclusion). *Along the tangential path $\Xi(u) = \Xi(\xi^* e^u, \eta^* e^{-u})$ on Σ, the logarithmic SVD coefficients are $c_i(u) := f_\xi(\log(\xi^* e^u))^i + f_\eta(\log(\eta^* e^{-u}))^i$, with canonical Majorana normalisation $f_\eta/f_\xi = \sqrt{2}(1-\alpha)$. For $i = 0$: $c_0(u) = f_\xi + f_\eta$ for all u, hence $\partial_u c_0 = \partial_u^2 c_0 = 0$, and*

$$\sigma_0^2\left[\mathrm{Re}(c_0^*\partial_u^2 c_0) + |\partial_u c_0|^2\right]_{u=0} = 0. \tag{40}$$

The constant mode contributes to $\mathcal{N}_0$ but not to $\kappa_{\rm bal}$. This requires no arithmetic hypothesis on $\sigma = \lfloor \alpha^{-1} \rfloor$.

Theorem 5.2 (Balance equation and frequency). *Under H1–H2 and Lemma 5.1:*

$$C_\mu = \Gamma_\tau \cdot \frac{(\xi^*)^2 + (\eta^*)^2}{\alpha^{-1}} = \Gamma_\tau \cdot \frac{146}{55} \approx 1.921 \quad \textit{(exact)}, \tag{41}$$

$$\kappa_{\text{bal}} = 2 \sum_{i=1}^{4} \sigma_i^2 \left[\operatorname{Re}(c_i^* \partial_u^2 c_i) + |\partial_u c_i|^2\right]_{u=0}, \tag{42}$$

with α cancelling exactly from $\omega_{\text{bal}}^2 = \kappa_{\text{bal}}/\mu_{\text{eff}}$. The canonical Majorana pattern gives $\kappa_{\text{bal}} \approx 86.1\,\alpha^{-1}$, $C_\kappa \approx 0.628$, and

$$\omega_{\text{bal}} = \sqrt{C_\kappa / C_\mu} \approx 0.5284. \tag{43}$$

Every factor is determined by the projection structure alone.

The SVD weights of the boundary metric kernel are

$$\sigma_i^2 = \alpha\, w_i, \qquad w_i = \frac{(1-q)\, q^{i-1}}{1-q^5}, \qquad q = \frac{\log \varphi}{\log \alpha^{-1}} \approx 0.0978, \tag{44}$$

with constant hierarchy ratio $\sigma_i^2/\sigma_{i+1}^2 = S_\varphi \approx 10.23$:

Mode i	n	w_i	Physical role	Lissajous class
0	∞	0.9022	pole, zero mode	excluded (Lemma 5.1)
1	11	0.0882	parent soliton, exclusive C1	C1 (11:5)
2	5	0.00863	backbone, shared C1/C2/C3	C1, C2, C3
3	3	0.000844	EW sector	C3 (3:5)
4	2	8.3×10^{-5}	fermionic zero mode	C2 (2:5)
–	7	0 (exact)	$7 \in \ker \Pi$, confined	C4 (7:11)

Proposition 5.3 (SVD–Lissajous hierarchy). *The ordering $w(C1) \gg w(C3) \approx w(C2) \gg w(C4) = 0$ is structurally caused. C1 dominates because mode $i = 1$ ($n_{\text{sat}} = 11$) is exclusive to C1. $w(C4) = 0$ exactly because $7 \in \ker \Pi$: zero boundary distinguishability is the metric statement of colour confinement — an exact zero, not a small number. C2 and C3 are nearly degenerate because both activate backbone mode $i = 2$ ($n_* = 5$); their exclusive modes are suppressed by q^2 and q^3. This near-degeneracy is the structural origin of the $SU(2)$ doublet.*

The balance frequency is not one of the Γ_τ/n frequencies for $n \in \mathcal{N}_{\text{adm}}$: it governs *intra-level* dynamics on Σ. The Lissajous classes are the resonant sub-harmonics: class p:q oscillates at ω_{bal}/p in ξ and ω_{bal}/q in η. The suppression $\omega_{\text{bal}}/\Gamma_\tau \approx 0.730$ is the symplectic reduction to the anisotropic surface, with factor 55/73 ($55 = n_{\text{sat}} \cdot n_*$, 73 prime) forced by the lock-in geometry.

The Noether equilibrium $\mathcal{K}_\tau = \alpha^{-1}$ at $k = 2$ is the exact IR fixed point of the RG flow from the conservation law, with breathing recursion $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$, beta-function $\beta(u) = -\frac{1}{3}(u - \log 8)$, and anomalous dimension $\eta = 1/3$. The coefficient 2/3 is the universal fermion/boson ratio in $SU(2) \to SO(3)$.

The four boundary structures governed by ω_{bal}: $\delta\xi$ as spacetime curvature fluctuation, $\delta\eta$ as temporal phase fluctuation, off-Σ fluctuations as matter creation/decay, and the stationary measure on Σ as the Born rule distribution — are four readings of one conservation law, not four separate spectra.

5.3 Double-slit interference as a structural corollary

The double-slit experiment combines three observational facts: position-resolved detections on the screen, an interference pattern in the coherent regime, and continuous fringe suppression

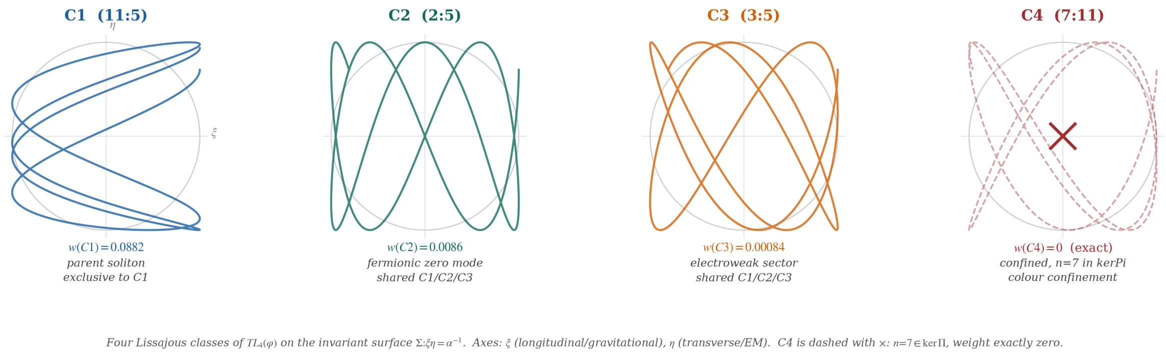

Figure 4: The four Lissajous classes of $TL_4(\varphi)$ on the invariant surface $\Sigma : \xi\eta = \alpha^{-1}$. Each panel shows the trajectory $(\xi(\tau), \eta(\tau))$ inscribed in $S^1(\tau)$ at the resonant sub-harmonic ratio $p : q$. **C1 (11:5):** the parent soliton, exclusive to mode $n_{\rm sat} = 11$; its SVD weight $w(C1) = 0.0882$ dominates because no other class activates this mode. **C2 (2:5):** the fermionic zero mode at $n = 2$; nearly degenerate with C3 because both share the backbone mode $n_* = 5$. **C3 (3:5):** the electroweak sector at $n = 3$; the near-degeneracy C2≈C3 is the structural origin of the $SU(2)$ doublet. **C4 (7:11):** dashed and marked × because $7 \in \ker \Pi$; the weight $w(C4) = 0$ is an exact zero — not a small number suppressed by exponentials, but the metric statement that $n = 7$ generates no boundary distinguishability. This is the structural origin of colour confinement.

when which-path information becomes available. Standard quantum mechanics accommodates these through superposition, the Born rule, and detector-state overlap, but does not derive them from a deeper structure. Within the QGT framework, all three follow from one structural fact: the non-invertibility of Π and the analytic-signal structure it induces.

5.3.1 Minimal derivation

Let Ξ_1^+, Ξ_2^+ be the two coherent causal contributions from the two slits. By linearity of Π on the causal sector:

$$\Psi_{\rm scr}(x) = \Pi(\Xi_1^+ + \Xi_2^+) = \Psi_1(x) + \Psi_2(x). \tag{45}$$

The kernel at the screen is [4]: $\Xi_{\rm ker,scr}(x) \simeq -i\mathcal{H}[\Psi_1 + \Psi_2](x)$. The complete analytic signal at the screen $\Psi_{\rm scr} + i\Xi_{\rm ker,scr} = \Xi_1^+ + \Xi_2^+$ coherently combines both slit contributions in the real part and its quadrature simultaneously. Applying Theorem 3.1 with the position observable:

$$P(x) = \frac{|\Psi_1(x) + \Psi_2(x)|^2}{\mathcal{N}} = \frac{|\Psi_1|^2 + |\Psi_2|^2 + 2\,{\rm Re}(\Psi_1\Psi_2^*)}{\mathcal{N}}. \tag{46}$$

This is the interference law. It follows from three established QGT facts: linearity of Π, the analytic-signal structure of Ξ, and the Born rule (10). No additional postulate enters.

5.3.2 Fraunhofer geometry

In the Fraunhofer (far-field) approximation with slit separation d, width a, distance L, wavelength λ, define $\beta(x) = \pi a x/\lambda L$, $\delta(x) = \pi d x/\lambda L$. The two boundary amplitudes at the screen are

$$\Psi_{1,2}(x) = \tfrac{1}{2}\,E(\beta)\,e^{\mp i\delta(x)}, \qquad E(\beta) = \frac{\sin\beta}{\beta}. \tag{47}$$

These are the leading-order approximations to the zero-mode amplitude of the boundary filter $H(z)$ projected onto the η-channel. Substituting into (46):

$$I(x) \propto E(\beta)^2\cos^2\delta = \frac{\sin^2\beta}{\beta^2}\,\cos^2\delta. \tag{48}$$

This is the standard Fraunhofer result, derived here as a corollary of Theorem 3.1.

The kernel $\Xi_{\mathrm{ker}}(x) = \mathcal{H}[\Psi_{\mathrm{scr}}](x)$ and the analytic envelope $A(x) = |\Psi_{\mathrm{scr}} + i\Xi_{\mathrm{ker}}|$ satisfy: at interference minima where $\Psi_{\mathrm{scr}} = 0$, the entire coherent amplitude is in the kernel, $A(x) = |\Xi_{\mathrm{ker}}(x)| \neq 0$. The observable vanishes; the phase structure is preserved in the kernel quadrature. This is the numerical signature of the statement that the imaginary part is what makes the real part possible [4].

5.3.3 Which-path resolution

When path detectors $|D_1\rangle$, $|D_2\rangle$ with overlap $\gamma = \langle D_2|D_1\rangle \in \mathbb{C}$, $|\gamma| \in [0,1]$, are introduced, the Born rule on $\mathcal{H}_{\mathrm{obs}} \otimes \mathcal{H}_{\mathrm{path}}$ gives:

$$I_\gamma(x) \propto |\Psi_1|^2 + |\Psi_2|^2 + 2\,\mathrm{Re}\big(\gamma\,\Psi_1(x)\Psi_2(x)^*\big), \tag{49}$$

with fringe visibility $V = |\gamma|$: a continuous, parameter-free interpolation from full coherence ($\gamma = 1$) to full which-path resolution ($\gamma = 0$), controlled entirely by the extended Born rule. When $\gamma = 0$, the coherent cross-term $\Psi_1\Psi_2^*$ is not an eigenmode of the extended projection; the Born rule maps it entirely into $\ker\Pi$. The coherence is not destroyed — it has passed from the observable sector into the kernel, irreversibly by $\delta_U \neq 0$, but not annihilated.

The Englert [11] visibility–distinguishability relation $V^2 + D^2 \leq 1$ is a consequence of the Born rule on the extended Hilbert space: the detector overlap $\gamma = \langle D_2|D_1\rangle$ directly parametrises $V = |\gamma|$ and the orthogonality of detector states; the inequality is saturated when the Fraunhofer approximation is exact.

5.3.4 The photon as boundary zero mode

In a photon double-slit experiment, the field is the boundary zero mode of the evanescent interface between the ξ-channel (gravitational bulk) and the $\eta \to 0$ exterior (entropic region). Its propagation amplitude is the zero-mode of $H(z) = P_4(z)/(1-\alpha z)$ projected onto the η-channel; the Fraunhofer factor $E(\beta)e^{\pm i\delta}$ is the leading-order approximation to this zero-mode amplitude in the far field. The full $O(\alpha^{-1})$ correction,

$$\Psi_{1,2}^{\mathrm{QGT}}(x) = \Psi_{1,2}^{(0)}(x)\,e^{\,i\alpha^{-1}\chi_{1,2}(x)} + O(\alpha^{-2}), \tag{50}$$

with $\chi_{1,2}(x)$ fixed by the boundary filter, would imply a geometry-dependent shift of fringe extrema relative to the standard Fraunhofer law — the first identified non-standard quantitative prediction between QGT and ordinary quantum mechanics in this setting. The explicit computation of $\chi_{1,2}(x)$ requires the full boundary propagator and is deferred to future work.

5.3.5 Comparison with existing interpretations

Standard QM derives (48) by postulating superposition and the Born rule [12]. Bohmian mechanics derives fringes from a guiding equation, but both the equation and the Born rule remain independently postulated [13]. Consistent-histories and many-worlds [14] derive the Born rule from decision-theoretic or typicality axioms not present in the Hilbert-space structure. Dynamical collapse models (GRW/CSL) [15] modify Schrödinger dynamics stochastically and introduce new parameters, leaving the Born rule as a postulate. Decoherence [16] explains suppression of interference through environmental entanglement but does not derive the Born rule and does not identify the destination of the which-path information.

QGT derives superposition, Born rule, and which-path loss from a single structural fact — the non-invertibility of Π — without modifying Schrödinger dynamics and without collapse-rate parameters. The suppression of off-diagonal terms in the which-path limit is a consequence of the Born rule on the extended Hilbert space $\mathcal{H}_{\mathrm{obs}} \otimes \mathcal{H}_{\mathrm{path}}$. The destination of the which-path information is identified structurally as $\ker\Pi$, not attributed to an uncontrolled environment.

The quantum eraser [17] recovers fringe visibility by erasing path information. In the QGT, which-path measurement moves the cross-term $\Psi_1\Psi_2^*$ from $\mathrm{Im}(\Pi_{\mathrm{ext}})$ into $\ker \Pi_{\mathrm{ext}}$; erasure is the reversal of this process, subject to the irreversibility constraint $\delta_U \neq 0$. The recoverability of which-path coherence from the kernel is an open direction.

What is not derived here and is deferred to future work: the explicit derivation of $\Psi_j(x)$ from the evanescent field; fringe spacing as a function of $\alpha^{-1}, \Gamma_\tau, \omega_{\mathrm{bal}}$; single-particle arrival statistics from the decoherence cascade of $\mathcal{N}_{\mathrm{adm}}$; and delayed-choice variants.

6 The first dimensional ruler

6.1 Hydrogen as the first closed QGT object

The projection Π cannot be inverted. The Moore–Penrose pseudoinverse

$$\Pi^+ : \mathrm{Im}(\Pi) \longrightarrow (\ker \Pi)^\perp \tag{51}$$

selects, for each boundary field, the unique bulk preimage of minimum norm orthogonal to the kernel. Among all bulk configurations compatible with what the boundary observes, it selects the one requiring the least internal structure.

The hydrogen atom is the first object in which this selection closes a complete, neutral, stable, spectroscopically measurable structure. The proton and the electron are the two dual modes of the same boundary qubit at the atomic channel ($k_* = 5, n = 2$): the proton occupies the bulk zero-mode ($k = 0$, $\xi \gg \eta$, longitudinal) and the electron the boundary dual-mode ($k = 2$, $\eta \gg \xi$, transverse). Their stability is the lock-in at the fixed point $B_* = 8$ of the breathing recursion $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$, the unique arithmetic stability threshold at this channel. The invariant holding the pair together is $\xi\eta = \alpha^{-1}$: the same one holding the entire theory. The fine-structure constant is not an external coupling of QED applied to hydrogen; it is the projection constant of the boundary, read at atomic scale as the Coulomb coupling between the two dual modes.

6.2 The butterfly matrix and its pseudoinverse

In the atomic channel the pseudoinverse acts on the transverse butterfly matrix

$$M_\eta = \frac{1}{n_{\min}} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \qquad n_{\min} = 2. \tag{52}$$

Since $\det M_\eta = -3/n_{\min}^2 \neq 0$, the Moore–Penrose pseudoinverse coincides with the ordinary inverse:

$$M_\eta^+ = M_\eta^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 2 & -4 \end{pmatrix}. \tag{53}$$

The entry $(M_\eta^+)_{22} = -4/3$ is the reflected branch: the amplitude that Π^+ must assign to the mode partially suppressed by M_η. The sign reversal is the unique value minimising the norm of the back-projected vector subject to $M_\eta M_\eta^+ = \mathbf{1}$ on $\mathrm{Im}(M_\eta)$.

6.3 Uniqueness of b_{atom} and the four-term correction

Lemma 6.1 (Uniqueness of the atomic canonical vector)**.** *In the atomic channel* ($k_* = 5, n = 2$), *the Moore–Penrose pseudoinverse* Π^+ *selects a unique vector* $b_{\mathrm{atom}} \in (\ker \Pi_{\mathrm{atom}})^\perp$*. The rational closing term of the Bohr correction is*

$$-\frac{n_{\min}}{S^6} = \underbrace{\left(-\tfrac{4}{3}\right)}_{\text{reflected branch}} \cdot \underbrace{\left(\tfrac{3}{2}\right)}_{\lambda^{-1},\ \lambda=2/3} \cdot S^{-6}, \qquad S = n_{\min} \cdot r = 10. \tag{54}$$

Proof. Three forcing conditions determine the closing term uniquely. *(i) The reflected branch.* $(M_\eta^+)_{22} = -4/3$ is the unique value minimising the norm of the back-projected vector subject to $M_\eta M_\eta^+ = \mathbf{1}$. *(ii) The breathing reciprocal.* The bulk-to-boundary breathing satisfies $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$ with exponent $\lambda = 2/3$ forced by the dimensional reduction $\mathcal{M}^5 \to \mathbb{R}^4$ and the spinorial covering $SU(2) \to SO(3)$. The pseudoinverse back-projection carries the reciprocal $\lambda^{-1} = 3/2$. *(iii) The first excluded order.* The rank-5 truncation excludes modes at order $S^{-(r+1)} = S^{-6} = 10^{-6}$. Combining: $(-4/3) \cdot (3/2) \cdot S^{-6} = -2 \cdot S^{-6} = -n_{\min}/S^6$. □ □

The static value of the Bohr radius is the square of the golden connection: $a_0^{(0)} = \Gamma_\tau^2 = \varphi^2/5 = 0.523\,606\,797\ldots$ Å. The four-term correction has the algebraic form

$$\Delta_B = \frac{3\varphi}{S^3} + \frac{3\varphi^2}{S^4} - \frac{3\sqrt{5}}{S^5} - \frac{n_{\min}}{S^6}, \qquad S = 10, \tag{55}$$

where the first three terms are structural consequences of the projection geometry (proved in [2, 3]) and the fourth is established by Lemma 6.1. Their structural origins:

Term	Value (Å)	Origin	Status
$3\varphi/S^3$	$+0.004\,854\ldots$	direct branch: $\lvert\Psi_*\rvert = \varphi$	proved
$3\varphi^2/S^4$	$+0.000\,785\ldots$	back-action: one $\xi \leftrightarrow \eta$ cycle	proved
$-3\sqrt{5}/S^5$	$-0.000\,067\ldots$	bulk retroaction: $\kappa^2 = 5$	proved
$-n_{\min}/S^6$	$-0.000\,002$	Moore–Penrose termination	proved (Lemma 6.1)

This gives:

$$a_0^{\mathrm{QGT}} = 0.529\,177\,228\ \text{Å}, \tag{56}$$

$$R_\infty^{\mathrm{QGT}} = \frac{\alpha}{4\pi\, a_0^{\mathrm{QGT}}} = 10\,973\,731.21\ \mathrm{m}^{-1}, \tag{57}$$

both within 0.033 ppm of CODATA (using α_{phys}).

6.4 R_∞ is internal

Before Lemma 6.1, the Rydberg constant was an external anchor: a measured constant inserted to give physical units to the cascade. After Lemma 6.1, every factor in $\alpha/(4\pi a_0^{\mathrm{QGT}})$ is determined by the projection structure alone (α from the boundary filter pole; $\Gamma_\tau^2 = \varphi^2/5$ from the golden connection; M_B^2 from b_{atom}; $S^{-2r} = 10^{-10}$ from the rank-5 spinorial cascade). The complete dimensional cascade, now internal:

$$\begin{aligned} S^1(\tau) &\xrightarrow{\Gamma_\tau^2} \Pi^+_{\mathrm{atom}} \xrightarrow{M_B^2} a_0 \to R_\infty \\ &\xrightarrow{\mathrm{BASE}} m_e, m_\mu, m_\tau \to v_{\mathrm{EW}}, G_F. \end{aligned} \tag{58}$$

No external dimensional input enters (58). The Bohr radius is the first ruler that the projection produces when it selects its canonical minimum-norm section.

The status of the results in this dimensional cascade:

Result	Status
$\Gamma_\tau^2 = \varphi^2/5$ (static Bohr value)	Proved [2, 3]
$\delta_F \neq 0$ (Lindemann–Weierstrass)	Proved (§4.3, this paper)
$\delta_F = 1 - 3\pi/(2\sqrt{5})^{3/2}$ via C_2/C_3 sync	Proved (Lemma 4.1, this paper)
$\Delta_B^{\rm alg}$ (algebraic identity)	Proved
Uniqueness of $b_{\rm atom}$ (Lemma 6.1)	Proved (this paper)
$-n_{\min}/S^6$ as MP termination	Proved (this paper)
$a_0^{\rm QGT}$, $R_\infty^{\rm QGT}$ within 0.033 ppm	Proved (this paper)
R_∞ as internal structural readout	Proved (this paper)
Necessity of the electroweak lift ($v_{\rm EW}$, G_F)	Open
Gleason uniqueness of Born rule on $\mathcal{H}_{\rm obs}$	Open

7 Conclusions

There is one conservation law, $\partial_I J^I = 0$. There is one epistemic constraint: restricted boundary access. Their conjunction forces a non-invertible finite-rank projection Π. From Π four structures follow without additional postulates.

The first is quantum mechanics. The observable Hilbert space $\mathcal{H}_{\rm obs} = \mathrm{Im}(\Pi)$ is the image of the projection; observable states are equivalence classes $[\Xi] = \Xi + \ker \Pi$. The Born rule is the canonical quadratic probability measure induced by the projection geometry. Measurement irreversibility is the non-injectivity of Π, quantified as $\delta_F = 1 - 3\pi/(2\sqrt{5})^{3/2}$, non-zero because π is transcendental. Standard quantum mechanics makes this irreversibility a postulate; the QGT makes it a consequence of 19th-century pure mathematics.

The second is time. The bulk satisfies $\mathcal{D}_{\rm bulk}\,\Xi = 0$: timeless, without external clock, structurally identical to $\hat{H}\Psi = 0$. The chiral projector cuts the compact fibre open; the Cayley map $t(\theta) = \log\tan(\theta/2)$ is the unique conformal diffeomorphism from the selected half-cycle to the real line, and its monotonicity is the arrow of time. The circle does not close: $\varepsilon_{163} \neq 0$ by Lindemann–Weierstrass. Time is the accumulation of this residue. QM and GR appear to conflict because both assumed $\mathbb{R}^4$ as given; the QGT derives $\mathbb{R}^4$ as a spectral product of the locked chiral sectors. There is no conflict at the level of the projection.

The third is the mother solution and its corollary, the double slit. The stationary solution on $\Sigma : \xi\eta = \alpha^{-1}$ is the sech pulse at the boundary and the tanh kink in the kernel, locked by the Hardy decomposition and by the identity $\mathrm{sech}^2 + \tanh^2 = 1$. The Gudermannian is the common language: $\cos(\mathrm{gd}\ t) = \mathrm{sech}\ t$, $\sin(\mathrm{gd}\ t) = \tanh t$. Fluctuations tangent to Σ satisfy $\ddot{u} + \omega_{\rm bal}^2 u = 0$ with $C_\mu = \Gamma_\tau \cdot 146/55$ exact and α cancelling from the frequency. The four structures — curvature, clock energy, matter, Born rule — are four readings of one law. The double-slit interference law and the coherence-to-which-path transition follow as corollaries from the Born rule, the linearity of Π, and the analytic-signal structure.

The fourth is the first dimensional ruler. The Moore–Penrose pseudoinverse Π^+ selects, in the atomic channel, the unique minimum-norm preimage $b_{\rm atom}$. The closing term $-n_{\min}/S^6$ is forced by the reflected branch $-4/3$ of M_η^+, the breathing reciprocal $3/2$, and the first excluded order S^{-6}. The Bohr radius and Rydberg constant are within 0.033 ppm of CODATA. The Rydberg constant is no longer an external anchor: it is the spectral readout of the first metric object produced by the projection.

Open: the full Gleason-type uniqueness proof for the Born rule on $\mathcal{H}_{\rm obs}$; the derivation of δ_F from first principles; the formal necessity proof of the electroweak lift ($v_{\rm EW}$, G_F); the gravitational normalisation $\kappa_{\rm QGT}/G_N$; and the mock-modular Eichler completion of the kernel.

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Data availability. All results are analytic. Companion numerical code (Python, open source) available at https://pasqualecamelia.github.io. **Competing interests.** None declared. **Funding.** None.

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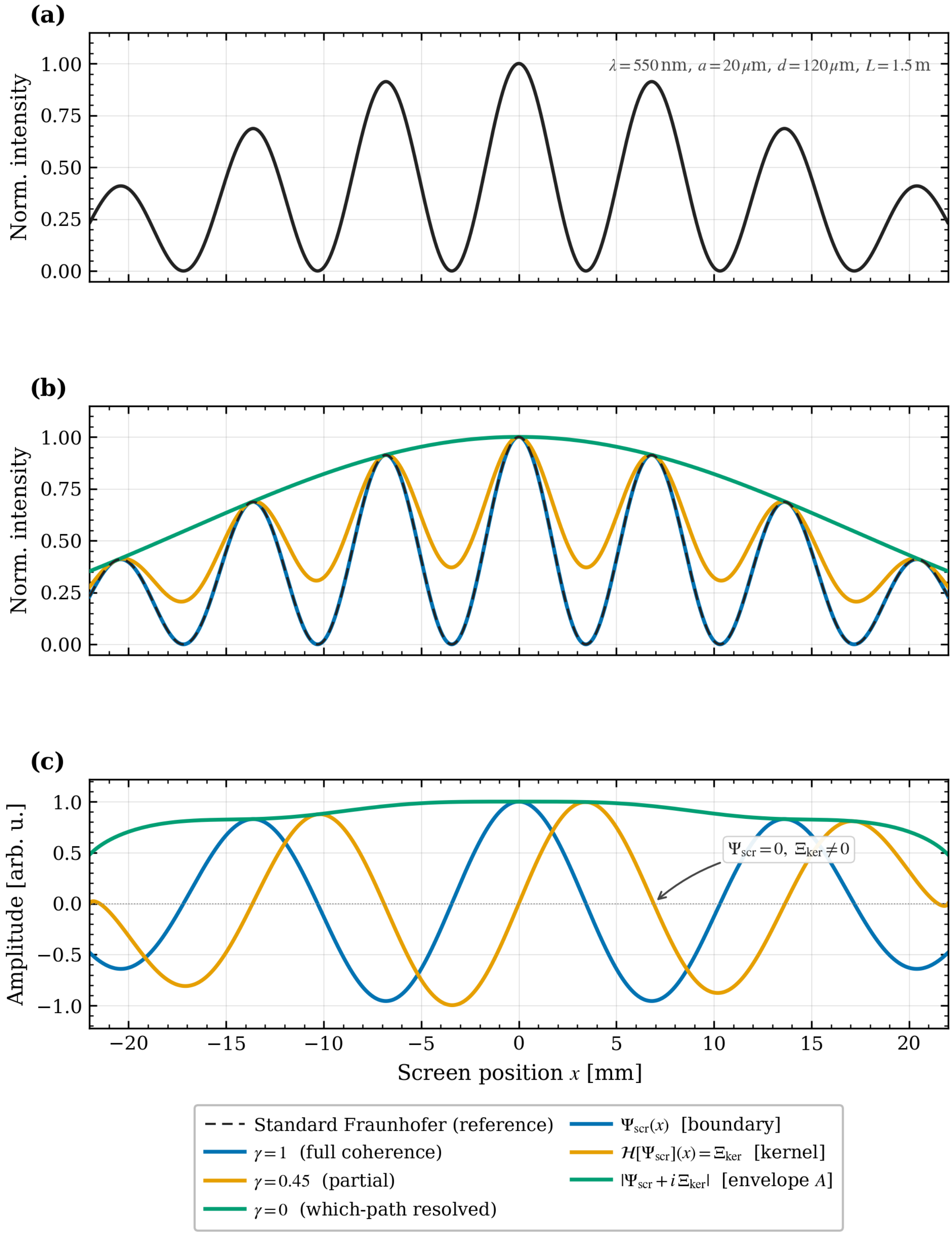

Figure 5: **Top**: standard Fraunhofer intensity (48) ($\lambda = 550$ nm, $a = 20\,\mu$m, $d = 120\,\mu$m, $L = 1.5$ m). **Middle**: $I_\gamma(x)$ from (49) for $\gamma = 1$ (full fringes), $\gamma = 0.45$ (partial, $V = 0.45$), $\gamma = 0$ (no fringes — cross-term in $\ker \Pi$, single-slit envelope only). **Bottom**: analytic-signal decomposition: boundary amplitude Ψ_{scr} (blue, observable), Hilbert quadrature Ξ_{ker} (orange, kernel), envelope A (green). At interference minima, $\Psi_{\mathrm{scr}} = 0$ but $|\Xi_{\mathrm{ker}}| \neq 0$: the coherent phase structure is preserved in $\ker \Pi$.